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ICT 4115: Cryptography and Cyber Law

Time: 3 Hours | Full Mark: 60

Attempt any five questions. Each question carries 12 marks.

1. (a) Determine whether each of the following forms a valid algebraic structure. Give proper justification. (4)
   i) Determine whether $(\mathbb{Z}_{15}, +, \times)$ forms a ring.
   ii) Determine whether $(\mathbb{Z}_{31}, +)$ and $(\mathbb{Z}_{31} \setminus \{0\}, \times)$ form Abelian groups?

   (b) Let $GF(2^2)$ be the finite field of order 4, constructed using the irreducible polynomial x^2+x+1 over $GF(2)$.
   i) Show that the nonzero elements of $GF(2^2)$ form a group under multiplication.
   ii) Verify whether the set of all nonzero elements of $GF(2^2)$ is cyclic.

   (c) Prove Fermat's Little Theorem and explain how it can be used to test for primality. Is the given integer 561 prime based on this test? Evaluate 5^{123} mod 175 using Fermat's Little Theorem. Show all steps.

2. (a) Use the Extended Euclidean Algorithm to find integers x and y satisfying the linear Diophantine equation $56x + 15y = 1$. Use a as the dividend, b as the divisor, q as the quotient, and r as the remainder, where $r = a - qb$.

   (b) Determine the following multiplicative inverses:
   i) $56^{-1} \pmod{15}$,
   ii) $43^{-1} \pmod{101}$.

   (c) State and prove the Chinese Remainder Theorem. Then solve the following system of congruences: $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$. (4)

3. (a) Consider the plaintext *INFORMATION SECURITY* and the Playfair key CRYPTOGRAPHY. Prepare the plaintext according to the standard Playfair rules, including inserting X between repeated letters when necessary and appending X if required to make the number of letters even. (6)
   i) Construct the complete 5×5 Playfair key matrix.
   ii) Encrypt the following plaintext: INFORMATION SECURITY
   iii) Decrypt the ciphertext to verify that the original prepared plaintext is recovered.

   (b) In the DES algorithm, a 64-bit plaintext block is divided into two 32-bit halves: L_0 and R_0. Given $R_0 =$ 0xF0F0F0F0 and round key $K_1 =$ 0x0F0F0F0F, compute the output of the first round's function $f(R_0, K_1)$ assuming XOR operation only. Then, find $L_1 = R_0$ and $R_1 = L_0 \oplus f(R_0, K_1)$; where $L_0 =$ 0xAAAAAAAA.

   (c) Find the value of x satisfaying $3^x \equiv 13 \bmod 17$. Show how you obtain the answer.

4. (a) Using Bézout's identity, determine whether the Diophantine equation $43x + 26y = 1$ has integer solutions. If it does, find one pair of integer solutions.

   (b) Given the input word: [0x23, 0xA7, 0x4C, 0x19]. Use the following partial AES S-box to perform the SubBytes transformation. Provide the resulting output word.

Row ↓ / Col →	3	4	7	9	A	C
1	6D			C6		
2	D4					
4		A1				2E
A			63		D2	

   (c) In AES encryption, apply the AddRoundKey step only. Given Input word = [0x1A, 0x2B, 0x3C, 0x4D] and Round key word = [0x55, 0x66, 0x77, 0x88]. Perform the XOR operation between the input word and the round key, and write the resulting output word. (5)

5. (a) Two users, Ayesha and Rahim, want to establish a shared secret using the RSA cryptosystem. They select the prime numbers $p = 5$ and $q = 11$. Using these values, perform the following: (5)

(i) Calculate the RSA modulus $n = pq$.

(ii) Calculate Euler's totient function $\phi(n) = (p-1)(q-1)$.

(iii) Choose $e = 3$ as the public exponent and verify that $\gcd(e, \phi(n)) = 1$.

(iv) Determine the private exponent d such that $ed \equiv 1 \bmod \phi(n)$

(v) If the message is $M = 9$, calculate the corresponding ciphertext C using $C = M^e \bmod n$.

Show all intermediate calculationjs clearly and state the resulting public key (e, n) and private key (d, n).

(b) Let the hash value of a message be $H(M) = 23$. An RSA system uses the private key $(d, n) = (43, 187)$. Generate the RSA digital signature by computing $S = H(M)^d \bmod n$. Show all intermediate steps and state the resulting digital signature. 56 (3) 4

(c) Aleya and Badol are using the Diffie–Hellman key exchange protocol. They agree on the following public parameters: $p = 23$ and $g = 5$. Aleya chooses the private key $a = 6$, while Badol chooses the private key $b = 15$. (4) 5

i) Compute Aleya's public key.

ii) Compute Badol's public key.

iii) Using the other party's public key, compute the shared secret key for both Aleya and Badol.

iv) Verify that both parties obtain the same shared secret.

6. (a) Consider an ElGamal digital signature scheme with the public parameters $p = 467, g = 2$. A sender chooses the private key $x = 127$. The hash value of the message is $H(M) = 100$. The sender selects a random integer $k = 213$, where $\gcd(k, p-1) = 1$. (4) 6

i) Calculate the public key $y = g^x \bmod p$. ~~372~~ 58

ii) Using the ElGamal signature algorithm, calculate the signature components r and s, where $r = g^k \bmod p, s \equiv k^{-1}(H(M) - xr) \pmod{p-1}$.

iii) Verify the signature using $g^{H(M)} \equiv y^r r^s \pmod{p}$. 2 ?. 5 5

(b) Consider the elliptic curve $y^2 \equiv x^3 + x + 1 \pmod{23}$. Verify whether $P = (3, 10)$ lies on the curve and compute $2P$ using the elliptic-curve point-doubling formula. Then compute $P + Q$ for $Q = (9, 7)$. (4)

(c) Suppose the elliptic curve $y^2 \equiv x^3 + 7x + 10 \pmod{37}$ has base point $G = (2, 5)$ of order $n = 19$. For private key $d = 9$, compute the public key $Q = dG$, generate the ECDSA signature (r, s) for $H(M) = 8$ using nonce $k = 3$, and verify the signature using Q. (4)

7. (a) A Kerberos ticket is issued at time $T = 1200$ seconds with a lifetime of 600 seconds. (3) 4

i) Determine the ticket expiration time.

ii) Suppose the Kerberos session key is calculated as $K = (K_c + K_s) \bmod 29$, where $K_c = 12$ and $K_s = 21$. Calculate the session key.

(b) Given the simplified MAC function $\text{MAC} = (M \times K) \bmod 17$, where $M = 8$ and $K = 5$, calculate the MAC value. (3) 4

(c) During an SSL/TLS handshake, suppose the client generates a random value $R_c = 25$ and the server generates $R_s = 17$. If the simplified session-key formula is $K = (R_c + R_s) \bmod 31$, calculate K. (3)

(d) An IoT device sends 1200 bytes of data every minute. Calculate the amount of data transmitted in 1 hour in bytes and KB. (3)

8. (a) Compare IPSec and SSL/TLS with respect to their security layers, typical applications, and advantages. Why is IPSec commonly used for VPNs? (3) 4

(b) Describe the major steps of the SSL/TLS handshake protocol. Explain how the handshake establishes: (4) 5

i) Server authentication

ii) Session keys

iii) Confidentiality

iv) Integrity

(c) Explain how a Distributed Denial-of-Service (DDos) attack works. Discuss any two techniques that can be used to detect or mitigate DDos attacks. (5)

CamScanner